

TIC 303470667

Eclipsing binary candidate dossier - Star Vetter

MARGINAL

BLS EPHEMERIS

Period 5.431910 d
Epoch (t0) 1657.9950 BTJD
Duration 0.1200 d
Depth 2.220 % (22,198 ppm)
SNR 57.0

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.03 (gate ≥ 0.50)
Robust depth sigma 38.6 (gate ≥ 7.0)
Odd/even z-score 2.61 (consistent)
Secondary eclipse not detected

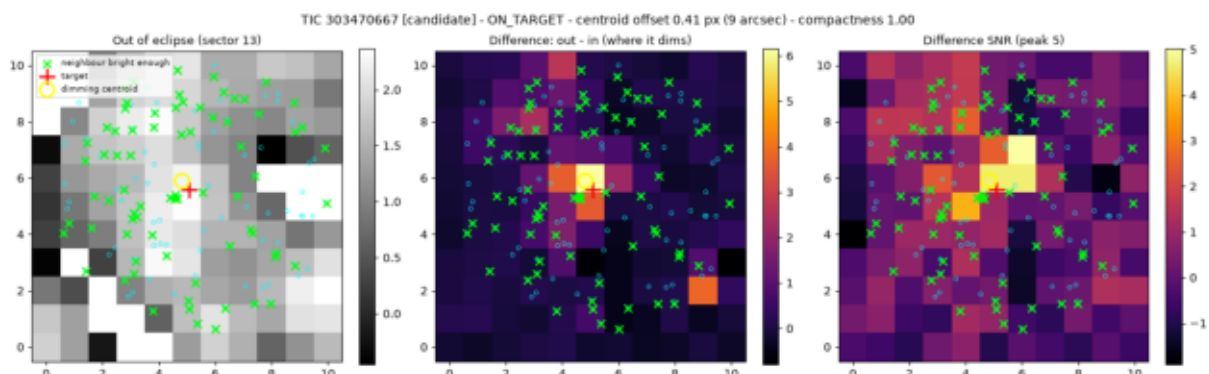
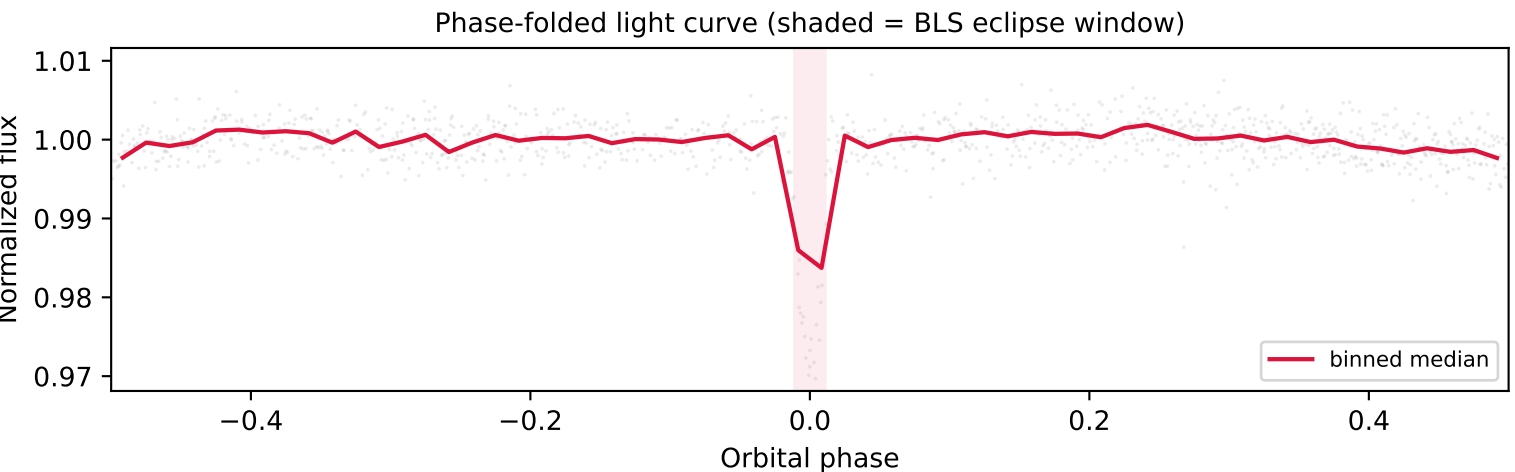
CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 13 light-curve dip in this sector: 33.5 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 5.02 (gate ≥ 5.0) Compactness 1.00 (gate ≥ 0.50)
Dimming centroid offset 0.41 px (9 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 1689765373
0.54 px away, dTmag +3.73 - farther from the dimming than the target



Caveat (MARGINAL): Pixel-check SNR is right at the gate (5.02 vs ≥ 5.0). A 150-trial false-alarm check on this star's own data found only a 3.3% chance of reaching this SNR at a random phase - the detection looks real, just faint. Weakest SNR margin of the confirmed set.